



Integrated Therapeutics

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URINALYSIS



Objectives

- ❑ Define Urinalysis and its relevance in diagnosis
- ❑ Discuss different examinations in urinalysis
- ❑ Understand and interpret urinalysis results

URINALYSIS

- Urinalysis is a cheap, fast and cost-effective diagnostic tool to have a look at what should or should not be present in the urine and can be a great indicator for disease or disorder.
- Provides valuable insights into renal function and systemic diseases.
- A complete urinalysis includes physical, chemical, and microscopic examinations.

Parameters to check during urinalysis

I. Physical examinations

- ❖ Colour and appearance
- ❖ Specific gravity
- ❖ PH
- ❖ Volume
- ❖ Odor

II. Chemical examinations

- ❖ Glucose
- ❖ Ketones
- ❖ Blood
- ❖ Protein
- ❖ Nitrites
- ❖ Leukocytes esterase
- ❖ Biluribin

III. Microscopic Examination of Urine

- ❖ RBCs
- ❖ WBCs
- ❖ Epithelial cells
- ❖ Casts
- ❖ Bacteria
- ❖ Yeasts
- ❖ Parasites
- ❖ Crystals

Colour and appearance

- Can be colourless or colourful, ranges from pale yellow to amber; indicates hydration status.
- The colour of urine can be changed by food, drug or the presence of blood. Drugs: Rifampin – reddish orange
propofol – green/blue
- Can be clear or cloudy
- Cloudy urine indicates that the presence of proteins or may be the indication of an infection.

Case

- A 45-year-old male, JD, presents with changes in urine color after starting Rifampin for tuberculosis; what is the specific change observed in his urinalysis, which shows dark orange/red urine that is clear
- A) Presence of hematuria
 - B) Dehydration
 - C) Side effect of medication
 - D) Liver dysfunction
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- Correct Answer: C) Side effect of medication

Specific gravity

- The following methods are used to test the specific gravity of urine:
 - ✓ Urinometer
 - ✓ Refractometer
 - ✓ Reagent strip
 - ✓ Weighing technique
- when concentration of urine increase specific gravity also increase
- Usually between 1.003 – 1.030, specific gravity is elevated because of the presence of sugar or protein.

cont....

- Increased USG is associated with:
 - glycosuria and
 - the syndrome of inappropriate antidiuretic hormone;
- Decreased USG is associated with
 - diuretic use
 - diabetes insipidus,
 - adrenal insufficiency, and
 - impaired renal function

pH

- Urinary pH can range from 4.5 to 8 but normally is slightly acidic (i.e., 5.5 to 6.5).
- reflects the function of the kidney in maintaining the acid-base balance
- Determination of urinary pH is useful in the diagnosis and management of UTIs,
- Alkaline urine in a patient with a UTI suggests the presence of a urea-splitting organism.

Urine Dipstick Method

- Urine dipstick method is a reagent strip test impregnated with chemicals called methyl red and bromothymol blue.
- The color change is interpreted by comparing the reference color chart supplies with the reagent strip.

cont...

Here's how it works:



- Dipping: The dipstick is dipped into a sample of urine.
- Reaction: The pads on the dipstick change color based on the presence and concentration of certain substances, such as glucose, protein, ketones, or blood.
- Reading Results: compare the colors on the dipstick to a color chart provided

Glucose

- Glucose normally, is almost completely reabsorbed in the proximal tubule.
- Glycosuria occurs when the filtered load of glucose exceeds the ability of the tubule to reabsorb. indicate:
 - diabetes mellitus,
 - Cushing's syndrome,
 - liver and pancreatic disease,

Ketones

- Normally not present in urine
- Ketonuria most commonly is associated with
 - uncontrolled diabetes,
 - occur during pregnancy,
 - carbohydrate-free diets, and
 - starvation

Blood

- The presence of blood in the urine is abnormal.
- Hematuria is associated with problems in the urinary tract such as
 - urological disease
 - damage at glomerulus membrane or
 - infection at tubules, bladder, urethra

Protein

- Normal urinary proteins include albumin, serum globulins, and proteins secreted by the nephron.
- Proteinuria is defined as urinary protein excretion of more than 150 mg per day.
- Elevated protein levels (proteinuria) can indicate
 - kidney damage,
 - glomerular issues,
 - high blood pressure, or other conditions affecting renal function.

Nitrites

- Nitrites are normally not present in urine.
- result when bacteria reduce urinary nitrates to nitrites,
- a positive dipstick nitrite test indicates that these organisms are present in significant numbers
- But, a negative result does not rule out UTI

Leukocytes (White Blood Cells)

- A small number of leukocytes may be present in normal urine.
- Elevated levels (leukocyturia) often indicate inflammation or infection, most commonly a UTI.
- This can be due to other inflammatory conditions.

Bilirubin

- Urine normally does not contain detectable amounts of bilirubin
- Bilirubinuria indicates further evaluation for liver dysfunction
 - biliary obstruction
 - or excessive red blood cell breakdown (hemolysis)

Case

- A patient presents with jaundice and dark urine. A urinalysis shows bilirubinuria. Which of the following is a most likely underlying condition?
 - a) Dehydration
 - b) Urinary tract infection
 - c) Kidney failure
 - d) Bile duct obstruction

Correct Answer: d) Bile duct obstruction

Casts

- Formed by precipitation of proteins, and aggregation of cells within the renal tubules
- Most of them dissociate in alkaline urine
- Presence of large number of hyaline casts may show possible damage of glomerular capillary membrane.

cont...

- Cellular casts contain
 - Epithelial cells: mostly seen in tubular degeneration
 - White blood cells: seen mostly during pyelonephrites conditions.
 - Red blood cells: usually seen in acute glomerulonephritis cases

Crystals

- Cellular Occurrence of certain abnormal urine crystals, such as cystine, Leucine, and Tyrosine, indicate the patient is in certain metabolic disorders

Bacteriuria

- Bacteriuria refers to the presence of bacteria in the urine
- asymptomatic Bacteriuria:
 - Presence of bacteria without symptoms
- Symptomatic Bacteriuria:
 - Presence of bacteria accompanied by symptoms such as:
 - Frequent urination
 - Painful urination (dysuria)
 - Urgency
 - Lower abdominal pain

Case

- A 28-year-old female presents with dysuria, increased urinary frequency, and urgency.
 - Urinalysis Results:
 - Color: Clear, pale yellow
 - Specific Gravity: 1.015
 - pH: 6.0
 - Nitrites: Positive
 - Leukocyte Esterase: Positive
 - Protein: Trace
 - Microscopic Examination: >10 white blood cells per high power field, few bacteria

The presence of nitrites and leukocyte esterase, along with elevated white blood cells, strongly suggests a urinary tract infection

Case

- A 45-year-old male with a family history of diabetes presents with polyuria, polydipsia, and fatigue.
 - Urinalysis Results:
 - Color: Amber
 - Specific Gravity: 1.030
 - pH: 5.0
 - Glucose: Positive
 - Ketones: Positive
 - Protein: Negative

The presence of glucose and ketones indicates uncontrolled diabetes mellitus, potentially leading to diabetic ketoacidosis.

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THANK YOU